

Interpret pie charts

Use the scaffold, show each step and ask for an adult check when marked.

Name: _____ Date: _____

1. Read the model: A quarter of a 240-pupil pie chart represents walkers. How many pupils walk? Copy or complete the first step.

2. In a survey of 60 pupils, 15 choose art. Find the pie-chart angle.

3. Find the mean of 12, 15, 18, 11 and 19.

4. Miro says: Miro divides by the largest value instead of the number of values when finding the mean. Is this correct? Explain.

Teacher answers and checking prompts

1. Read the model: A quarter of a 240-pupil pie chart represents walkers. How many pupils walk? Copy or complete the first step.
Answer 60 pupils.
2. In a survey of 60 pupils, 15 choose art. Find the pie-chart angle.
Answer 90 degrees.
3. Find the mean of 12, 15, 18, 11 and 19.
Answer 15.
4. Miro says: Miro divides by the largest value instead of the number of values when finding the mean. Is this correct? Explain.
Answer Add every value, count how many values there are, then divide the total by that count.

Interpret pie charts

Work independently. Show a complete method and check at least one answer.

Name: _____ Date: _____

1. A quarter of a 240-pupil pie chart represents walkers. How many pupils walk?

6. Check this result using an inverse, estimate or known fact: Find the mean of 12, 15, 18, 11 and 19.

2. In a survey of 60 pupils, 15 choose art. Find the pie-chart angle.

7. Prove or explain your reasoning: Dataset A has mean 24 over 8 values. Dataset B has total 207 over 9 values. Which mean is greater?

3. Find the mean of 12, 15, 18, 11 and 19.

8. Identify and correct this misconception: Miro divides by the largest value instead of the number of values when finding the mean.

4. Solve this independently and show every stage: A quarter of a 240-pupil pie chart represents walkers. How many pupils walk?

9. Write a complete sentence explaining how you checked one answer.

5. Use a second method or representation for: In a survey of 60 pupils, 15 choose art. Find the pie-chart angle.

10. Write and solve a similar question to: Find the mean of 12, 15, 18, 11 and 19.

Teacher answers and checking prompts

1. A quarter of a 240-pupil pie chart represents walkers. How many pupils walk?
Answer 60 pupils.
2. In a survey of 60 pupils, 15 choose art. Find the pie-chart angle.
Answer 90 degrees.
3. Find the mean of 12, 15, 18, 11 and 19.
Answer 15.
4. Solve this independently and show every stage: A quarter of a 240-pupil pie chart represents walkers. How many pupils walk?
Answer 60 pupils.
5. Use a second method or representation for: In a survey of 60 pupils, 15 choose art. Find the pie-chart angle.
Answer Expected result: 90 degrees. Method or representation may vary.
6. Check this result using an inverse, estimate or known fact: Find the mean of 12, 15, 18, 11 and 19.
Answer Expected result: 15. A valid check must be shown.
7. Prove or explain your reasoning: Dataset A has mean 24 over 8 values. Dataset B has total 207 over 9 values. Which mean is greater?
Answer Dataset A: 24; Dataset B: 23, so A is greater. The explanation must show why the method works.
8. Identify and correct this misconception: Miro divides by the largest value instead of the number of values when finding the mean.
Answer Add every value, count how many values there are, then divide the total by that count.
9. Write a complete sentence explaining how you checked one answer.
Answer Answer should name an inverse, estimate, boundary, known fact or equivalent representation.
10. Write and solve a similar question to: Find the mean of 12, 15, 18, 11 and 19.
Answer Answers vary. The new question must use the same mathematical structure and include a correct solution.

Interpret pie charts

Prove, compare or generalise. Write enough reasoning for another pupil to follow.

Name: _____ Date: _____

1. Prove or explain your reasoning: Dataset A has mean 24 over 8 values. Dataset B has total 207 over 9 values. Which mean is greater?

6. Create a new example that tests the same idea as: Find the mean of 12, 15, 18, 11 and 19.

2. Prove the answer to this question, rather than only calculating it: A quarter of a 240-pupil pie chart represents walkers. How many pupils walk?

7. Find a second method or representation. Compare its efficiency with your first method.

3. Solve this in two different ways and compare them: In a survey of 60 pupils, 15 choose art. Find the pie-chart angle.

8. Write one always, sometimes or never statement about today's learning and justify it.

4. Work backwards from the answer to reconstruct the question: 15.

9. Create a believable incorrect solution. Identify the exact step where the reasoning breaks.

5. Explain why this reasoning fails: Miro divides by the largest value instead of the number of values when finding the mean.

10. Change one value or condition in a question. Explain what changes in the solution and what stays the same.

Teacher answers and checking prompts

1. Prove or explain your reasoning: Dataset A has mean 24 over 8 values. Dataset B has total 207 over 9 values. Which mean is greater?

Answer Dataset A: 24; Dataset B: 23, so A is greater. The explanation must show why the method works.
2. Prove the answer to this question, rather than only calculating it: A quarter of a 240-pupil pie chart represents walkers. How many pupils walk?

Answer Expected result: 60 pupils. Proof or justification must match the lesson structure.
3. Solve this in two different ways and compare them: In a survey of 60 pupils, 15 choose art. Find the pie-chart angle.

Answer Expected result: 90 degrees. Two valid methods and a comparison are required.
4. Work backwards from the answer to reconstruct the question: 15.

Answer One valid reconstruction is: Find the mean of 12, 15, 18, 11 and 19. Other equivalent questions are acceptable.
5. Explain why this reasoning fails: Miro divides by the largest value instead of the number of values when finding the mean.

Answer Add every value, count how many values there are, then divide the total by that count.
6. Create a new example that tests the same idea as: Find the mean of 12, 15, 18, 11 and 19.

Answer Answers vary. The example and solution must preserve the mathematical structure.
7. Find a second method or representation. Compare its efficiency with your first method.

Answer Answers vary. Comparison should refer to accuracy, number structure and clarity.
8. Write one always, sometimes or never statement about today's learning and justify it.

Answer Answers vary. A valid example and counterexample should support the classification.
9. Create a believable incorrect solution. Identify the exact step where the reasoning breaks.

Answer Answers vary. The error must be mathematically relevant and the correction must be precise.
10. Change one value or condition in a question. Explain what changes in the solution and what stays the same.

Answer Answers vary. The explanation must distinguish the mathematical structure from the changed value.